

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 02

Construction Contracts

Course	BECM 3101 - Construction and Project Management I
Lecture deck	2.0 - Construction Contracts (47 PDF pages/slides)
Question bank	Term-final examinations, 2016-2024 (25 pages)
Method	Question wording and answer-content matched to specific lecture slides, regardless of the original Section A/B label

RESULT

This is a major examination lecture: 27 past-question links were found across all nine years. The strongest recurring areas are Lump Sum vs GMP, Cost Reimbursement + Unit Price, Schedule of Rates, and the legal definition of a construction contract.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 02 directly supports a large portion of the contract-related examination questions. The original exam sections change by year, so this report maps by subject matter rather than by the printed Section A/B heading.

47	27	1	13	13	328
Slides reviewed	Question links	Exact match	Direct matches	Partial matches	Mapped marks

Classification	Meaning
Exact	The requested definition/list appears essentially as asked and can be answered directly from the cited slides.
Direct	Most or all answer content is present; the student must apply, compare, select or combine the material.
Partial	Only one component or supporting concept is present; another lecture/source is required for a complete answer.

Slide-numbering note

The mapping uses the PDF page/slide count with the title page as Slide 1. The deck's small printed slide number is generally one less because the title page is unnumbered. Example: PDF Slide 5 is printed as slide number 4 in the deck.

Highest-priority slide blocks

Priority	Slides	Topic	Reason
1	9, 11-17, 44-46	Lump Sum vs GMP	Repeated comparison/selection questions in 2019, 2020, 2021 and 2023; also supports 2022 and 2024.
2	20-21, 29-30	Unit Price + Cost Reimbursement suitability	Near-identical paired questions in 2019, 2021 and 2024.
3	24-26	Schedule of Rates	Scenario-based questions in 2018 and 2022.
4	5-6	Legal definition and contract outline	Legal-definition questions recur in 2016, 2019, 2020, 2021, 2022 and 2023.

2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed meaning. 'QB p.' identifies the page in the uploaded 2016-2024 question bank.

Examinations 2016-2018

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Sec A Q1(a)	10	1	Define construction management from different points of view; discuss the characteristics of a project plan.	Partial	3	Slide 3 supplies a basic definition of construction management. The different viewpoints and project-plan characteristics are not developed in this deck.
2016	Sec B Q5(a)	15	2	Define construction contract; briefly discuss the contract outline; explain the statement 'Never had contract with good person.'	Partial	5-6	Slides 5-6 directly cover the legal/payment definitions and contract outline. The quoted statement is not explained.
2016	Sec B Q5(b)	10	2	What conditions are required to build a good construction contract?	Partial	6-9	The deck provides the contract outline, responsibility, terms, rights and contract-selection factors, but does not present a formal 'good contract conditions' checklist.
2016	Sec B Q5(c)	10	2	What are the factors affecting and negotiating contract types?	Partial	7-9	Slides 7 and 9 cover factors affecting contract type and selection. Contract negotiation is not separately explained.
2017	Sec B Q5(a)	10	5	What is construction management? Write down the objectives of construction management.	Exact	3-4	The definition and the stated objectives appear directly on Slides 3-4.
2017	Sec B Q5(b)	15	5	A single tendered price is given for a high-rise commercial building. Identify the appropriate contract and state its main aspects, advantages and disadvantages.	Direct	9, 11-16	The scenario indicates a lump-sum contract. Slides 11-16 cover its main aspects, advantages, disadvantages and selection context.
2018	Sec A Q1(c)	13	7	A schedule of work items without quantity is prepared by the client. Identify the appropriate contract and discuss its main aspects, advantages and disadvantages.	Direct	24-26	This is a Schedule of Rates contract. Slides 24-26 cover the exact identifying feature and the requested discussion.
2018	Sec B Q5(a)	12	8	Differentiate Construction Management from Project Management and define a project with an example.	Partial	3	Only the construction-management definition is present. Project definition, example and CM-PM comparison are outside this deck.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2019-2021

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2019	Sec B Q5(a)	10	10	Define construction contract from a legal point of view; state the objectives of cost estimation for different stakeholders.	Partial	5	The legal definition is exact on Slide 5. Cost-estimation objectives belong to another lecture.
2019	Sec B Q5(b)	15	10	State the suitability of a cost-reimbursement contract and briefly explain the main aspects of a unit-price contract.	Direct	20-21, 29-30	Suitability is covered on Slides 29-30; unit-price main aspects are covered on Slides 20-21.
2019	Sec B Q5(c)	10	10	Choose either a Lump Sum contract or a Guaranteed Maximum Price (GMP) contract; justify the selection with examples.	Direct	9, 11-17, 44-46	The deck explains both contract types and gives risk/price examples. Slide 9 supports the selection rationale.
2020	Sec B Q4(a)	12	13	Define construction contract from a legal point of view; choose Lump Sum or GMP and justify the choice with examples.	Direct	5, 9, 11-17, 44-46	The legal definition and both contract alternatives are directly supported. The final choice requires applying the selection factors.
2020	Sec B Q4(b)	18	13	State the suitability of cost-reimbursement contracts and describe the construction-project development process from the contractor's perspective.	Partial	29-30	The cost-reimbursement suitability is directly covered. Contractor-side project development is absent.
2021	Sec A Q1(a)	12	15	Differentiate Construction Management and Project Management; state the SMART rule for project objectives.	Partial	3	Slide 3 supports only the construction-management definition. The comparison and SMART rule are not in this lecture.
2021	Sec B Q5(a)	8	16	Define construction contract from a legal point of view; state the objectives of cost estimation for different stakeholders.	Partial	5	The contract definition is exact. The cost-estimation part is not covered.
2021	Sec B Q5(b)	15	16	State the suitability of cost-reimbursement contracts and explain the main aspects of a unit-price contract.	Direct	20-21, 29-30	Both requested topics appear directly in the mapped slide ranges.
2021	Sec B Q5(c)	12	16	Choose either Lump Sum or GMP, explain the reason for selection and give an example.	Direct	9, 11-17, 44-46	The deck supplies the characteristics and examples needed for a justified comparison.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2022-2023

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2022	Sec B Q5(a)	10	19	Define construction contract; briefly describe the objectives of cost estimation for different stakeholders.	Partial	5	Only the construction-contract definition is covered in this lecture.
2022	Sec B Q5(b)	12	19	A schedule of work items without quantity is prepared by the client. Identify the contract and discuss its main aspects, advantages and disadvantages.	Direct	24-26	The identifying condition and all requested Schedule of Rates features are contained on Slides 24-26.
2022	Sec B Q5(c)	13	19	When should a cost-reimbursement contract be recommended? Briefly explain Lump Sum and GMP contracts.	Direct	11-17, 27-32, 44-46	All three contract groups are covered, including suitability, main features, risk and price behavior.
2023	Sec A Q1(a)	10	20	Define construction management; state the main criteria for considering a proposed industrial building as a project.	Partial	3	The construction-management definition is present. Project criteria are outside the lecture.
2023	Sec B Q5(a)	10	21	Define construction contract from a legal point of view; state the objectives of cost estimation for different stakeholders.	Partial	5	Slide 5 gives the legal definition; the cost-estimation objectives are not included.
2023	Sec B Q5(b)	12	21	A schedule of work items with quantities is prepared by the A/E. Identify the appropriate contract and state its main aspects, advantages and disadvantages.	Direct	20-23	This describes a BOQ/Unit-Price contract. Slides 20-23 contain the exact requested material.
2023	Sec B Q5(c)	13	21	Choose either Lump Sum or GMP, justify the choice and explain with examples.	Direct	9, 11-17, 44-46	The comparison is directly supported by the two contract sections and selection factors.

2. Detailed Year-wise Question Mapping (continued)

Examinations 2024-2024

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2024	Sec B Q5(b)	12	24	State the suitability of cost-reimbursement contracts and briefly explain the main aspects of a unit-price contract.	Direct	20-21, 29-30	This is the same high-frequency pairing seen in 2019 and 2021; both parts are directly covered.
2024	Sec B Q7(b)	12	25	Actual costs plus a fixed fee are reimbursed, with a pre-agreed maximum limit and client benefit from savings. Identify the contract and explain its main aspects, advantages and disadvantages.	Direct	36-39, 44-46	The scenario points to a GMP/cost-plus-fixed-fee arrangement with a ceiling. Slides 36-39 and 44-46 provide the relevant risk, fee and maximum-price features.
2024	Sec B Q8(c)	17	25	Using given costs and change orders, determine which contract gives the least benefit among Lump Sum, Unit Price, Cost Plus Percentage, Cost Plus Fixed Fee, Target Cost and GMP.	Partial	10-17, 20-23, 33-46	The deck explains the contract behavior needed for interpretation, but it does not provide the numerical comparison procedure or formulas required for the full solution.

3. Repeated Question Patterns

The following patterns should guide exam preparation. A repeated cluster means that the wording is identical or the tested decision is substantially the same.

#	Pattern	Years/questions	Slides	Exam insight
1	CM definition / CM vs PM	2016 Q1(a); 2017 Q5(a); 2018 Q5(a); 2021 Q1(a); 2023 Q1(a)	Slides 3-4	5 appearances. Only 2017 Q5(a) is fully covered; comparison/project criteria require another lecture.
2	Legal contract definition + cost-estimation objectives	2019 Q5(a); 2021 Q5(a); 2022 Q5(a); 2023 Q5(a)	Slide 5	4 near-identical questions. Learn the legal definition here; obtain cost-estimation objectives from the costing lecture.
3	Cost reimbursement suitability + Unit Price main aspects	2019 Q5(b); 2021 Q5(b); 2024 Q5(b)	20-21, 29-30	3 near-identical questions with strong direct coverage.
4	Lump Sum vs GMP selection	2019 Q5(c); 2020 Q4(a); 2021 Q5(c); 2023 Q5(c)	9, 11-17, 44-46	4 repeated comparison/choice questions. This is the most important decision-based cluster.
5	Schedule of Rates identification	2018 Q1(c); 2022 Q5(b)	24-26	2 scenario questions built around the phrase 'work items without quantity'.
6	Combined Cost Reimbursement, Lump Sum and GMP	2022 Q5(c)	11-17, 27-32, 44-46	One broad integrative question; excellent revision test for the major contract groups.
7	Multi-contract numerical comparison	2024 Q8(c)	10-17, 20-23, 33-46	New numerical/application style. Contract concepts are present, but calculation rules must come from class notes or another source.

Quick recognition rules

Question clue	Likely contract	Slides
Single fixed/tendered price for a clearly defined job	Lump Sum	11-17
A/E supplies work items and estimated quantities; contractor enters unit rates	BOQ / Unit Price	20-23
Schedule of work items without quantities; repair/maintenance/urgent work	Schedule of Rates	24-26
Actual costs reimbursed plus a fee; vague scope, emergency, concurrent design	Cost Reimbursement	27-32
Agreed target with bonus/penalty or sharing of savings	Target Cost	40-43
Cost-plus arrangement with an assured maximum/ceiling	GMP	44-46

4. Slide-to-Question Reverse Index

Use this page while studying the lecture: after finishing a slide block, immediately practice the listed past questions.

Slides	Topic	Past questions	Study note
3-4	Construction management definition and objectives	2016 Q1(a); 2017 Q5(a); 2018 Q5(a); 2021 Q1(a); 2023 Q1(a)	Mixed - one exact, four partial
5	Construction contract: legal and payment definitions	2016 Q5(a); 2019 Q5(a); 2020 Q4(a); 2021 Q5(a); 2022 Q5(a); 2023 Q5(a)	High repetition; often paired with material from another lecture
6	Contract outline	2016 Q5(a), Q5(b)	Direct support for outline; partial for broader 'good contract' conditions
7-9	Variation in contract types and factors influencing choice	2016 Q5(c); 2017 Q5(b); 2019 Q5(c); 2020 Q4(a); 2021 Q5(c); 2023 Q5(c)	Core selection/justification support
11-17	Lump Sum	2017 Q5(b); 2019 Q5(c); 2020 Q4(a); 2021 Q5(c); 2022 Q5(c); 2023 Q5(c); 2024 Q8(c)	Very high priority
19	Admeasurement overview	Supports 2018 Q1(c), 2022 Q5(b), 2023 Q5(b)	Useful bridge to BOQ and Schedule of Rates
20-23	BOQ / Unit Price	2019 Q5(b); 2021 Q5(b); 2023 Q5(b); 2024 Q5(b); 2024 Q8(c)	Very high priority
24-26	Schedule of Rates	2018 Q1(c); 2022 Q5(b)	Learn the no-quantity clue
27-32	Cost Reimbursement	2019 Q5(b); 2020 Q4(b); 2021 Q5(b); 2022 Q5(c); 2024 Q5(b)	Very high priority
33-35	Cost Plus Percentage	2024 Q8(c)	Numerical comparison support only
36-39	Cost Plus Fixed Fee and risk sharing	2024 Q7(b); 2024 Q8(c)	Important for the latest exam style
40-43	Target Cost with variable fee	2024 Q8(c)	Numerical comparison support only
44-46	Guaranteed Maximum Price	2019 Q5(c); 2020 Q4(a); 2021 Q5(c); 2022 Q5(c); 2023 Q5(c); 2024 Q7(b); 2024 Q8(c)	Very high priority

5. Recommended Study Plan and Coverage Gaps

A practical order is to study by repeated exam pattern rather than by slide order.

Step	Study block	Slides	Practice immediately
1	Legal definition, outline and contract-choice factors	5-9	2016 Q5(a-c), 2020 Q4(a)
2	Lump Sum and GMP comparison	11-17, 44-46	2019 Q5(c), 2021 Q5(c), 2023 Q5(c)
3	Unit Price and Cost Reimbursement	20-23, 27-32	2019 Q5(b), 2021 Q5(b), 2024 Q5(b)
4	Schedule of Rates	24-26	2018 Q1(c), 2022 Q5(b)
5	Cost-plus variants, Target Cost and latest application questions	33-46	2024 Q7(b), Q8(c)

Topics mentioned in questions but not fully covered by Lecture 02

Gap	Affected questions	Action
CM vs PM comparison, project definition/criteria and SMART rule	2018 Q5(a); 2021 Q1(a); 2023 Q1(a)	Map these to a project-management fundamentals lecture.
Characteristics of a project plan	2016 Q1(a)	Use the project-planning lecture.
Objectives of cost estimation for owner/designer/contractor	2019 Q5(a); 2021 Q5(a); 2022 Q5(a); 2023 Q5(a)	Use the costing/estimating lecture; do not try to answer this part from Lecture 02.
Contract negotiation and the 'good person' statement	2016 Q5(a), Q5(c)	Check class notes or another professional-issues lecture.
Contractor-perspective project development process	2020 Q4(b)	Use the contractor/project-development lecture.
Numerical formulas for comparing contract outcomes	2024 Q8(c)	The deck gives concepts only; obtain the worked calculation method separately.

Final recommendation

Memorize the recognition clues first, then build comparison tables for Lump Sum, Unit Price, Schedule of Rates, Cost Reimbursement, Cost Plus, Target Cost and GMP using: price certainty, scope certainty, risk allocation, flexibility, incentive, payment basis, suitability, advantages and disadvantages.